

Long-run θ follows the unbalanced accumulation path

Historical directional object

$$\theta_{\Gamma,t}^{\text{LR}} = \left. \frac{d \ln Y_t^p}{d \ln K_t^{\text{cap}}} \right|_{\Gamma^{\text{LR}}}$$

Γ^{LR} is the actual or estimated long-run direction in the $(K^{\text{ME}}, K^{\text{NR}})$ capital space. The derivative therefore evaluates the capacity payoff of the accumulation process the economy actually follows.

Identification verdict

Annual observations reveal local path slopes; the accumulation window supplies the long-run regime object. Neither requires balanced growth.

Two empirical resolutions

Local: $\theta_{\Gamma,t} = g_t^p / g_{K,t}$

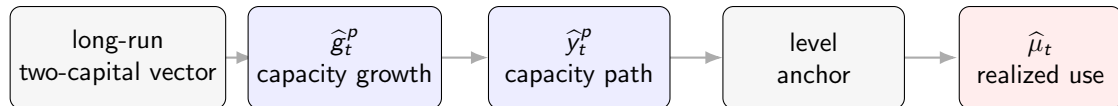
Long-run window: $\Theta_{\Gamma,w}^{\text{LR}} = \frac{\sum_{t \in w} g_t^p}{\sum_{t \in w} g_{K,t}}$

Analytical source: A03 aggregate transformation benchmark; S33 heterogeneous-capital recovery ledger

Observed output still separates capacity from realized use

Accounting separation

$$y_t = y_t^p + \log \mu_t$$



The residual remains an admissibility object

Residual stationarity admits the long-run capacity relation. Only the reconstructed and anchored y_t^p determines the level of μ_t .

Analytical source: Binding distribution-conditioned identification sequence

Persistent distribution selects technique before installation

Long-run technique choice

$$q_t^* = \arg \max_q [a(q) - q\pi_t^{\text{LR}}]$$

$$a'(q_t^*) = \pi_t^{\text{LR}} = 1 - \omega_t^{\text{LR}}$$

- The persistent distributive state selects the technique before installation.
- The selected technique governs the ME/NR bundle of the new increment.

Separate the horizons

$$\omega_t = \omega_t^{\text{LR}} + \omega_t^{\text{SR}}$$

ω_t^{LR}

selects the technique of new accumulation

ω_t^{SR}

changes realized output and utilization without repricing installed capacity

Causal implication

Distribution conditions the new capital increment; it cannot revalue the physical capacity of yesterday's stock.

Fixed technique does not freeze stock composition

Technique fixes the marginal bundle

$$\rho_t^I(q_{t-1}^*) = \frac{dK_t^{ME}}{dK_t^{NR}}$$

The installation rule fixes the composition of the new investment increment. It does not impose the composition of the inherited stock.

The stock ratio changes unless both coincide

$$d\tau_t = \frac{dK_t^{ME}}{K_t^{ME}} - \frac{dK_t^{NR}}{K_t^{NR}}$$

A fixed marginal bundle generally produces $d\tau_t \neq 0$ because its ME/NR ratio differs from the stock inherited from previous installations.

The category error is precise

Fixed technique is a causal timing restriction. Fixed stock composition is a balanced-growth restriction. Treating them as equivalent removes the mechanization mechanism from the object being identified.

The two-capital surface separates plant and machinery payoffs

Composition-mediated long-run differential

$$dy_t^P = \theta^{scale} dk_t^{NR} + \theta_t^\tau d\tau_t, \quad \tau_t = k_t^{ME} - k_t^{NR}$$

Equivalent component-capital representation

$$dy_t^P = \underbrace{(\theta^{scale} - \theta_t^\tau)}_{\theta_t^{NR}} dk_t^{NR} + \underbrace{\theta_t^\tau}_{\theta_t^{ME}} dk_t^{ME}$$

θ_t^{NR} : plant-envelope contribution to capacity formation.

θ_t^{ME} : machinery and technique contribution to capacity formation.

Identification verdict

The decomposition does not replace aggregate θ . It identifies the structural contributions from which aggregate θ is constructed along the accumulation path.

Capital growth and potential-output growth share one path

Exact capital coordinate

$$\begin{aligned}K_t^{cap} &= K_t^{ME} + K_t^{NR} \\k_t^{cap} &= k_t^{NR} + \ln(1 + e^{\tau_t}) \\dk_t^{cap} &= dk_t^{NR} + s_t^{ME} d\tau_t\end{aligned}$$

Different magnitudes, one movement

$g_{K,t} = \Delta \ln K_t^{cap}$
is productive-capital growth.

$g_t^P = \Delta \ln Y_t^P$
is the potential-output response generated along that same capital movement.

Why the distinction must remain

Imposing $g_t^P = g_{K,t}$ would impose $\theta = 1$ by construction. The ratio compares the response with its capital-stock base; it does not introduce a second accumulation path.

Analytical source: D10 capital identity; A03 constant-price capital aggregation

The accumulation direction produces aggregate θ

One capital path, two growth rates

$$g_t^P = \theta^{scale} g_t^{NR} + \theta_t^\tau (g_t^{ME} - g_t^{NR}), \quad \theta_{\Gamma,t} = \frac{g_t^P}{g_{K,t}} = \theta^{scale} + \left(\theta_t^\tau - s_t^{ME} \theta^{scale} \right) \frac{d\tau_t}{dk_t^{cap}}.$$

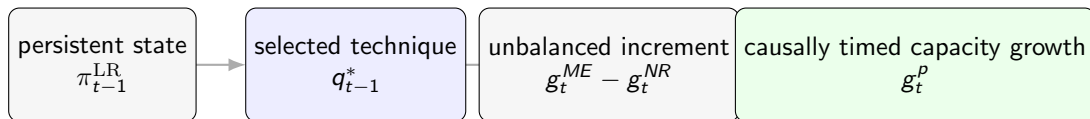
two-capital gradient $\longrightarrow$ unbalanced accumulation direction $\longrightarrow$ aggregate $\theta_{\Gamma,t}$

Balanced accumulation is only a special benchmark

Only when $d\tau_t = 0$ does the formula reduce to θ^{scale} . Along the historical unbalanced path, composition enters the elasticity tested against unity.

Analytical source: A03 aggregate ratio; S33 component-capital bridge; v4 directional identity audit

Embodiment changes timing, not the accumulation direction



Recursive capacity reconstruction

$$g_t^P = \theta^{scale} g_t^{NR} + \theta_{t-1}^T (g_t^{ME} - g_t^{NR}), \quad y_t^P = y_{t-1}^P + g_t^P$$

Embodiment verdict

Lagging the technique payoff prevents current distribution from repricing installed capital. It does not suppress the ME/NR direction through which the new technique enters the stock.

Analytical source: S33 causal order and composition-mediated Golden-Age reconstruction

The affine profit-share form requires composition restrictions

Map the estimated wage-share state

$$\pi_t^{\text{LR}} = 1 - \omega_t^{\text{LR}}$$

$$\theta_t^\tau = \psi + \lambda_\omega \tilde{\omega}_{t-1}^{\text{LR}} \equiv \psi_\pi + \lambda_\pi \pi_{t-1}^{\text{LR}}$$

Aggregate reduced form

$$\theta_{\Gamma,t}^{\text{LR}} = \theta^{\text{scale}} + \left[\psi_\pi + \lambda_\pi \pi_{t-1}^{\text{LR}} - s_t^{\text{ME}} \theta^{\text{scale}} \right] r_t^{\text{LR}}$$

$$r_t^{\text{LR}} \equiv \frac{d\tau_t}{dk_t^{\text{cap}}}$$

The restriction must be visible

The governing form $\theta_t = \theta_1 + \theta_2 \pi_t$ follows only when r_t^{LR} and the relevant composition share are fixed over the marginal comparison, or when an explicit approximation imposes that closure. Otherwise aggregate θ remains path-dependent.

Analytical source: Binding affine lock; composition-mediated specification; v4 methodological correction

Unity classifies the combined path elasticity

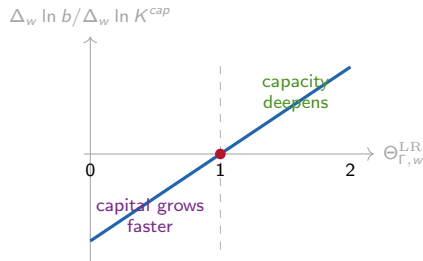
Accumulation-window object

$$\Theta_{\Gamma,w}^{\text{LR}} = \frac{\sum_{t \in w} g_t^p}{\sum_{t \in w} g_{K,t}}$$

$$\Delta_w \ln b = (\Theta_{\Gamma,w}^{\text{LR}} - 1) \Delta_w \ln K^{\text{cap}}$$

Conditional on $\sum_{t \in w} g_{K,t} > 0$:

- $\Theta_{\Gamma,w}^{\text{LR}} < 1$: capital grows faster than the capacity it builds;
- $\Theta_{\Gamma,w}^{\text{LR}} = 1$: capacity productivity is unchanged;
- $\Theta_{\Gamma,w}^{\text{LR}} > 1$: capacity grows more than proportionally.



Window rule

Contraction years enter the cumulative path; classification requires positive cumulative capital growth for the window.

Analytical source: Harrodian accounting identity and A03 aggregate transformation object

Regimes require windows; realized use still requires μ

Window interval	Accumulation verdict	Interpretive boundary
$U_w^\Theta < 1$	below-one capital-overaccumulation tendency	capital expands faster than the potential capacity it produces
$L_w^\Theta \leq 1 \leq U_w^\Theta$	indeterminate	the data do not identify a side of the Harroddian benchmark
$L_w^\Theta > 1$	above-one capacity-deepening tendency	excess productive capacity additionally requires unused or increasingly unused capacity

Utilization after path reconstruction

$$\Delta \ln \mu_t = \Delta y_t - g_t^p = \Delta y_t - \theta_{\Gamma,t} g_{K,t}$$

Final analytical verdict

$\Theta_{\Gamma,w}^{\text{LR}}$ classifies capacity creation. Anchored μ_t determines its use: above-one capacity deepening becomes excess productive capacity only when realization fails.

Analytical source: Constitutional productive-capacity corridor; binding capacity-utilization sequence

A1. Identification fails when any path gate fails

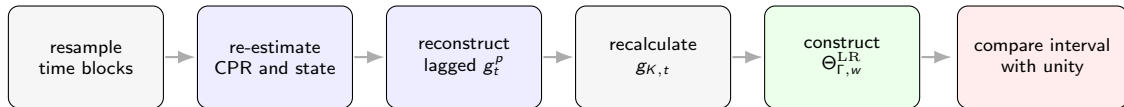
Gate	Pass condition	Failure classification
state order	k_t^{NR} , τ_t , and distributive conditioning are admissible long-run objects	mixed-order or polynomial-risk specification
rank	scale and composition effects are separately identified	mechanization collapses into the scale direction
residual	$\hat{\varepsilon}_t \sim I(0)$	no stable long-run capacity surface
timing	lagged persistent distribution weights the new ME/NR increment	contemporaneous repricing of installed capacity
aggregation	$K_t^{cap} = K_t^{ME} + K_t^{NR}$ and the growth denominator is reproduced	capital-path denominator is incoherent
anchoring	one explicit benchmark fixes the level of y_t^p	μ_t remains underidentified
accumulation	$\sum_{t \in w} g_{K,t} > 0$	unity has no threshold interpretation for the window
inference	the interval for $\Theta_{r,w}^{LR}$ is economically admissible	no structural regime verdict

Rank has an economic meaning

The data must contain independent variation in plant-envelope scale and ME/NR composition. Without it, the gradient cannot be separated and the path elasticity cannot be structurally decomposed.

Analytical source: Binding specification fork; A03/A05 measurement and promotion gates

A2. Inference must rebuild the entire unbalanced path



Required protocol

1000 moving-block draws with block length 4 years. Every draw re-estimates the coefficient and generated-state chain.

Current inference status

protocol defined; interval not yet estimated. The existing roughness-attribution bootstrap is not a regime interval, so this deck makes no numerical historical regime claim.

Analytical source: Required generated-state inference; 13 claims and 10 identities validated